Lecture 11

Decoherence: the unread environment

Objectives. *An environment can record alternatives without anyone reading its record. The resulting channel suppresses interference in a part while the combined state can remain pure. We compute this for a qubit and for position-dependent random kicks, then distinguish the small-separation and resolved-collision regimes before estimating two physical decoherence times.*

- Compute a dephasing factor from environmental record overlap and distinguish a reduced mixture from a selected outcome.
- Prove the Gaussian position-kick channel's kernel factor with the correct infinite-dimensional scope.
- Check the regime and units of a scattering estimate for a dust grain and for a model molecule in thermal radiation.

11.1 A record changes a part's interference

Let the system be a qubit and the environment have a fixed unit ready vector e. For an environment unitary U_E take the joint coupling

$$U = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes U_E.$$

Its two orthogonal system sectors show that U is unitary. A pure input $\psi = a\,|0\rangle + b\,|1\rangle$ gives

$$\Psi = a\,|0\rangle \otimes e + b\,|1\rangle \otimes U_E e.$$

The total remains a unit vector. Put $c = \langle e, U_E e\rangle$. L6's local pairing gives the reduced operator

$$|a|^2\,|0\rangle\langle 0| + a\bar{b}\,\langle U_E e, e\rangle\,|0\rangle\,\langle 1| + \bar{a}b\,\langle e, U_E e\rangle\,|1\rangle\,\langle 0| + |b|^2\,|1\rangle\langle 1|\,.$$

In particular the $|0\rangle\,\langle 1|$ coefficient is $\bar{c}\,a\bar{b}$; the rank-one trace identity is $\mathrm{Tr}(|u\rangle\,\langle v|) = \langle v, u\rangle$. By linearity the reduced map is

$$\begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longmapsto \begin{pmatrix} \rho_{00} & \bar{c}\rho_{01} \\ c\rho_{10} & \rho_{11} \end{pmatrix}. \tag{11.1}$$

The conjugate on the upper entry follows from our antilinear-first inner product. Cauchy–Schwarz gives $|c| \le 1$. Orthogonal environmental records have $c = 0$ and remove the local coherence completely; parallel records have $|c| = 1$ and change only its phase. We call c the environmental overlap, or *decoherence factor*; the upper coherence uses $\bar{c}$ and its magnitude is multiplied by $|c|$. It measures how much interference survives when the record is ignored. The map is a channel by L7's initially separate environment construction.

For a qubit environment initially $|0\rangle$, choose $U_E = e^{-i\theta\sigma_y/2}$. Then $c = \cos(\theta/2)$. At $\theta = \pi$ the records are orthogonal. For $a = b = 1/\sqrt{2}$ the joint output is a Bell vector, while the system becomes $\mathbb{1}_2/2$. Its purity is $\mathrm{Tr}\,\rho_1^2 = 1/2$ although the joint purity is one. No stochastic choice of 0 or 1 occurred in this unitary calculation. At $c = 0$ the map is $\rho \mapsto \sum_{j=0}^{1} |j\rangle\langle j|\,\rho\,|j\rangle\langle j|$, the unread Lüders channel of L10. Reading an environmental record and selecting its outcome would be the extra operation described by an instrument in L10.

A mixture prepared by an ignored classical random label is often called *proper*. The same density operator obtained by restricting an entangled pure state is called *improper*, in d'Espagnat's terminology promised in QM-I, Lecture 11, after the half of a singlet as a mixture. These words concern the preparation and the larger state, not two kinds of density operator. Local bounded observations cannot distinguish them. Nor does a diagonal reduced operator identify one privileged ensemble: L2 already exhibited many preparations of one state. A global test can distinguish the entangled state from a global incoherent mixture; L12 computes it.

Self-check 11.1

1. For equal qubit amplitudes, what is the probability of $+x$ in terms of c?
2. At $c = 0$, does the partial trace specify which pointer outcome occurred?

11.2 Position dephasing as a channel

On $L^2(\mathbb{R})$ take $\Lambda_{\mathrm{dec}} > 0$ with units length^{-2} time^{-1}. For $t > 0$ define the probability density

$$\gamma_t(k) = \frac{e^{-k^2/(4\Lambda_{\mathrm{dec}}t)}}{\sqrt{4\pi\Lambda_{\mathrm{dec}}t}}, \qquad V(\hbar k) = e^{ikQ}.$$

Here k has inverse-length units and $\hbar k$ is the momentum kick. The variance of k is $2\Lambda_{\mathrm{dec}}t$. Define the output by its bounded expectations,

$$\mathrm{Tr}(\Phi_t(\rho)A) = \int \gamma_t(k)\,\mathrm{Tr}(\rho e^{-ikQ}Ae^{ikQ})\,dk. \tag{11.2}$$

L7 proved that this scalar rule constructs a CP, trace-preserving map: this is (7.4) with its scalar density equal to γ_t and unitary family $\mathcal{W}(k) = V(\hbar k)$. At $t = 0$ define $\Phi_0 = \mathrm{id}$ separately; there is no ordinary density γ_0. Thus a symbolic integral of operators is not the existence argument for this channel.

Proposition 11.1 (Gaussian suppression of position coherence)**.** *For $t \geq 0$, the kernel ρ_t of $\Phi_t(\rho_0)$ satisfies*

$$\rho_t(x, x') = e^{-\Lambda_{\mathrm{dec}}t(x-x')^2}\rho_0(x, x'). \tag{11.3}$$

For general trace-class inputs kernels are interpreted in $L^2(\mathbb{R}^2)$, hence almost everywhere, rather than as prescribed pointwise values.

Proof. For a rank-one wave packet, conjugation by e^{ikQ} multiplies its kernel by $e^{ik(x-x')}$. The Gaussian Fourier integral is

$$\int \gamma_t(k)e^{ikd}\,dk = e^{-\Lambda_{\mathrm{dec}}td^2}.$$

For completeness put $\zeta = \Lambda_{\mathrm{dec}}t > 0$ and call the integral $J(d)$. Since $k\gamma = -2\zeta\gamma'$, differentiation and integration by parts give $J'(d) = -2\zeta d J(d)$, with $J(0) = 1$; Gaussian tails justify the operations by finite cutoffs. Solving gives the displayed factor. To connect that integral

with the scalar definition of the channel, first take input φ and testing vectors f, g in C_c^∞. Use $A = |g\rangle\langle f|$ in (11.2), cut k off at $\pm K$, and exchange the continuous iterated Riemann integrals on the resulting compact box. The omitted tail is at most $\|f\|\,\|\varphi\|^2\,\|g\| \int_{|k|>K} \gamma_t(k)dk$, by Cauchy–Schwarz. Taking $K \to \infty$ gives

$$\langle f, \Phi_t(|\varphi\rangle\langle\varphi|)g\rangle = \iint \bar{f}(x)e^{-\Lambda_{\mathrm{dec}}t(x-x')^2}\varphi(x)\overline{\varphi(x')}g(x')\,dx\,dx'.$$

S4 extends the tests and input to L^2: the left side is continuous in the input by the uniform bound $|\langle V\varphi, AV\varphi\rangle - \langle V\xi, AV\xi\rangle| \le \|A\|\,(\|\varphi\| + \|\xi\|)\,\|\varphi - \xi\|$ for unitary V. The right side has the same continuity by the product-kernel L^2 bound. Matrix elements separate operators, proving the rank-one identification. Linearity now gives the kernel for finite spectral sums.

To pass to every positive trace-class operator, its spectral kernels $\sum_j p_j\varphi_j(x)\overline{\varphi_j(x')}$ converge in $L^2(\mathbb{R}^2)$, since those product vectors are orthonormal and $\sum_j p_j^2 < \infty$. A square-integrable kernel defines a bounded operator of norm at most its L^2 norm, by Cauchy–Schwarz followed by integration. The operator defined by this kernel limit agrees with ρ by the operator-norm convergence of positive spectral tails. Multiplication by the Gaussian factor has norm at most one on kernel space, so the transformed kernels also converge there. The channel of a positive omitted input tail is positive with the same small trace, hence has operator norm at most that trace. Its operator limit is consequently $\Phi_t(\rho)$, identifying the kernel limit. Complex linearity extends the conclusion to L1's trace-class span. At $t = 0$ it is the identity. ∎

The kernel factors multiply, so $\Phi_s\Phi_t = \Phi_{s+t}$. This explicit infinite-dimensional channel semigroup did not use L8's finite GKSL theorem. Position probabilities stay unchanged: each e^{ikQ} commutes with $E_Q(I)$, so (11.2) gives $\mathrm{Tr}(\Phi_t(\rho)E_Q(I)) = \mathrm{Tr}(\rho E_Q(I))$. A diagonal of a general L^2 kernel is not determined by its equivalence class; this bounded-event identity is the general population statement. For smooth kernels one can also read it from $x = x'$ in (11.3).

For finite-rank operators with Schwartz eigenvectors, differentiating the explicit factor produces the formal expression

$$\dot{\rho} = -\Lambda_{\mathrm{dec}}[Q,[Q,\rho]], \qquad L_1 = \sqrt{2\Lambda_{\mathrm{dec}}}\,Q,$$

with free motion neglected. The double commutator's kernel is $(x - x')^2\rho(x, x')$. This identifies the differential rule on these test kernels; it is not a claim that every trace-class input is differentiable, or that this test class has been proved a generator core. The channel is defined on all trace class by (11.2). The spectral shift rule gives $e^{-ikQ}E_P(I)e^{ikQ} = E_P(I - \hbar k)$, so bounded momentum-event probabilities obey the convolution rule of (10.4). For finite input momentum variance its ordinary variance addition gives $2\hbar^2\Lambda_{\mathrm{dec}}t$, by the physical kick variance.

Self-check 11.2

1. At separation d, what time reduces the kernel factor to e^{-1}?
2. Why does the transformed spectral kernel sequence converge in L^2, and why is the positive output tail norm at most its trace?

11.3 Which scattering regime does the Gaussian model describe?

The Gaussian channel is mathematically valid for every separation. Its identification with a scattering experiment has a restricted regime. The following kinetic derivation is *sketched*,

following Schlosshauer, §3.1–3.4 pp. 119–138. It assumes a massive object with negligible recoil, independent stationary scattering events, and negligible free motion during the coherence comparison. A symmetric incident bath removes directed phase drift. These are physical assumptions, not consequences of CP.

Let Γ be the total collision rate and q the momentum-transfer component along the separation. In one event the coherence gains a factor $\chi(d) = \langle e^{iqd/\hbar} \rangle$ from the transfer distribution. During a short interval, the probabilities of no collision and one collision are $1 - \Gamma\, dt$ and $\Gamma\, dt$ to first order. Thus

$$\dot{\rho}(x, x') = \Gamma(\chi(x - x') - 1)\rho(x, x').$$

For a symmetric distribution with finite second moment and $|qd/\hbar| \ll 1$ at typical transfers,

$$\chi(d) \simeq 1 - \frac{\langle q^2 \rangle d^2}{2\hbar^2}, \qquad \Lambda_{\text{dec}} = \frac{\Gamma \langle q^2 \rangle}{2\hbar^2}.$$

This is the quadratic small-separation law. If one uses the total three-dimensional transfer magnitude in an isotropic bath, its component second moment is one third of that total; the factor must be converted.

At separations resolved by the scattered particles, the overlap factor in the cited isotropic model averages close to zero. The decay rate then approaches Γ, rather than increasing without limit as d^2. The wavelength language is an order-of-magnitude criterion; the precise small parameter is transfer times separation divided by $\hbar$. Saturation is an asymptotic property of the specified scattering model, not a universal monotonicity theorem for every conceivable kick distribution. One must not extrapolate the quadratic coefficient to a macroscopically resolved separation just because its Gaussian channel remains positive.

In the controlled-qubit model, the interaction preserves the system basis 0, 1 and makes its superpositions entangle with distinct records. A *pointer basis* here consists of vectors $|j\rangle$ for which $U(|j\rangle \otimes e) = |j\rangle \otimes e_j$: they remain products while their superpositions can entangle with records. Position-dependent scattering similarly favours localised packets or coarse position sectors. Exact delta functions are not pointer states in this Hilbert space. The widths, stability and competition with free spreading depend on the interaction and timescale; decoherence does not select one universal basis for every Hamiltonian.

Self-check 11.3

1. When should the rate scale as d^2, and when as the total collision rate?
2. Which assumption removes the term linear in momentum transfer from the rate estimate?

11.4 A dust grain in air, and a model molecule in light

Take air at 300 K, pressure $p \simeq 1.01 \cdot 10^5$ Pa, and mean molecular mass $m \simeq 4.8 \cdot 10^{-26}$ kg. The ideal-gas and Maxwell estimates give

$$n = \frac{p}{k_B T} \simeq 2.45 \cdot 10^{25}\,\text{m}^{-3}, \qquad \bar{v} = \sqrt{\frac{8 k_B T}{\pi m}} \simeq 470\,\text{m s}^{-1}.$$

Specify the grain's *diameter* as 10 μm, so its radius is $a = 5\,\mu$m. With geometric effective cross section πa^2,

$$\Gamma \simeq n \bar{v} \pi a^2 \simeq 9 \cdot 10^{17}\,\text{s}^{-1}, \qquad \tau_{\text{dec}} \simeq \Gamma^{-1} \simeq 10^{-18}\,\text{s}.$$

At a separation $d = 1\,\mu$m, the thermal wavelength $h/\sqrt{2\pi m k_B T} \simeq 1.9 \cdot 10^{-11}\,$m is much smaller than d. This is the resolved, approximately saturated regime. The quadratic formula would give an unjustified much shorter time. The stated time is a characteristic stationary scattering-model estimate, not a measurement of its initial microscopic transient. It depends on ideal-gas flux, a heavy stationary grain and the chosen effective cross section; changing radius to 10 μm would multiply the rate by four. Units check directly: $n\bar{v}\pi a^2$ is inverse time.

For a contrasting quadratic-regime example, take Schlosshauer's model large molecule of radius 10 nm in room-temperature photon radiation. Table 3.1, p. 135, estimates

$$\Lambda_{\mathrm{dec}} \simeq 10^6\,\mathrm{cm^{-2}s^{-1}} = 10^{10}\,\mathrm{m^{-2}s^{-1}}.$$

Choose $d = 10\,$nm. Then $\Lambda_{\mathrm{dec}} d^2 \simeq 10^{-6}\,\mathrm{s^{-1}}$ and $\tau_{\mathrm{dec}} \simeq 10^6\,$s, about twelve days. Both the object and the separation are small compared with the roughly 10 μm thermal-photon wavelength. The *small separation* justifies the quadratic rate; the *small object* justifies the Rayleigh cross section used to estimate its coefficient. They are distinct conditions. The table uses a dielectric-sphere approximation with contrast of order one and a strong radius/temperature dependence $\Lambda_{\mathrm{dec}} \propto a^6 T^9$. It is not a measured universal coefficient for every molecule, nor the radius of a fullerene. These different reservoirs and objects need not have comparable times.

The physics

Hornberger et al., *PRL* **90**, 160401 (2003), observed loss of fullerene interference with increasing background-gas pressure (doi:10.1103/PhysRevLett.90.160401). This supports collisional decoherence, with an apparatus-specific scattering model. It is separate from our model dielectric sphere's thermal-photon estimate. Environmental coherence loss changes an interference contrast; seeing that contrast disappear alone does not establish selection of one global outcome.

Self-check 11.4

1. Why is the air estimate based on Γ rather than $\Lambda_{\mathrm{dec}} d^2$?
2. For the same quadratic coefficient, what happens to the model molecule's time if its separation doubles while remaining in the small-separation regime?

Why are position superpositions difficult to observe?

A part coupled to an initially separate, unread environment evolves by a channel. Records suppress its interference while the joint state can remain pure. In the stated scattering regime the small-separation decay rate is $\Lambda_{\mathrm{dec}} d^2$; resolved collisions instead give a rate near the total collision rate. Position probabilities stay fixed in this pure-dephasing approximation. The reduced mixture supplies no unique selected outcome; that remaining question is the measurement problem.

Summary. Environmental overlap determines the qubit's coherence. L7's scalar construction gives a genuine Gaussian kick channel, whose kernel factor is proved in the appropriate L^2 sense. Quadratic diffusion and resolved-collision saturation describe different physical regimes. The two estimates illustrate why assumptions and scales precede the numerical rate. Proper and improper refer to preparation context; neither local density statistics nor diagonalisation by itself chooses an outcome.

Self-check answers.

11.1 1. $(1 + \operatorname{Re} c)/2$. 2. No; it only gives the unread probabilities.

11.2 1. $1/(\Lambda_{\mathrm{dec}} d^2)$ for $d \neq 0$. 2. The Gaussian multiplier has norm at most one on L^2; the output tail is positive and L1 bounds its operator norm by its trace.

11.3 1. Below the transfer-resolution scale, and well above it in the stated scattering model, respectively. 2. Symmetry of the transfer distribution; a directed bath can add a phase/drift.

11.4 1. The separation is far larger than the resolving wavelength. 2. The time becomes one quarter as large.

Sources. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, pp. 88–93, 119–138; saturation equations (3.45)–(3.48) p. 129, Rayleigh coefficient (3.66)–(3.67) p. 134 and Table 3.1 p. 135. The air calculation uses the saturated collision rate, not an extrapolation of Table 3.2's quadratic values. The explicit channel proof uses L1, L6 and L7.